

13. (a) Explain what is meant by a carbon-neutral fuel. [3]

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- (b) Ethane and an unknown alkane were burned in oxygen.

It follows from the equation below that one volume of ethane produced two volumes of carbon dioxide and three volumes of water vapour.



10 cm³ of the unknown alkane C_xH_y burned according to the following equation.



The total volume of carbon dioxide and water vapour produced was 20 cm³ more than the original volume of C_xH_y and oxygen. All volumes were measured at the same temperature and pressure.

- (i) State the volumes of carbon dioxide and water vapour produced on burning 10 cm³ of the unknown alkane in terms of x and y . [1]

Volume carbon dioxide = cm³

Volume water vapour = cm³

- (ii) Calculate the value of y . [2]

$y =$

END OF PAPER

